

DevOps Series

Ansible Deployment of the Aerospike NoSQL Database

In this 13th article in the DevOps series, we will install and set up an Aerospike NoSQL database cluster.



Aerospike is a free and open source, distributed NoSQL database written in C. It is primarily designed for high throughput and low latency applications. It is a key-value store optimised for Flash storage devices and runs on *nix platforms. It can also be used as an in-memory database residing entirely on RAM. Following the CAP (Consistency, High Availability, Partition Tolerance) theorem, at present, the database can be operated in AP mode. The database functions in three layers — the data storage layer, the clustering and distribution layer, and the client layer. As in many distributed databases, the server supports rolling upgrades and failover mechanisms in the cluster.

A number of client libraries for various programming language environments like C, Java, Go, Python, Node.js, etc, are available to create applications that can communicate with the Aerospike server. The basic data types that are supported are integer, string, bytes, double, list, map, GeoJSON and blobs (language serialised). Support for user defined functions (UDF) also exists, using the Lua programming language. The Aerospike tooling ecosystem has a number of connectors for the Spring framework, Hadoop, Kafka, etc. The Aerospike Management Console (AMC) is a Web-based application used to monitor the Aerospike cluster. The Aerospike server was first launched in 2012, and is released under the Affero General Public License.

GNU/Linux

A host system running Ubuntu 16.04.3 LTS (Xenial) with Vagrant and VirtualBox is used to launch three CentOS 6.9 virtual machines, on which the Aerospike server will be installed and set up. The Vagrant file to launch the three nodes is shown below:

```
BOX_IMAGE = "bento/centos-6.9"
NODE_COUNT = 3
```

```
Vagrant.configure("2") do |config|
  (1..NODE_COUNT).each do |i|
    # Create Aerospike server node
    config.vm.define "as#{i}" do |node|
      node.vm.box = BOX_IMAGE
      node.ssh.forward_agent = true
      # https://github.com/mitchellh/vagrant/issues/4967
      node.ssh.insert_key = false
      node.ssh.username = 'vagrant'
      node.ssh.password = 'vagrant'
      node.vm.network :private_network, ip: "10.0.0.#{i + 10}"
      node.vm.hostname = "as#{i}"
      node.vm.provider "virtualbox" do |vb|
        vb.name = "Aerospike-server-#{i}"
        vb.memory = "2048"
      end
    end
  end
end
```

The three instances are named 'as1', 'as2' and 'as3', and are assigned private IP addresses that can be accessed from the host system. The machines are brought up using the following command:

```
$ vagrant up
```

The version of Ansible used on the host system is 2.2.0.0. A project directory structure is created to store the Ansible playbooks and inventory, as follows:

```
ansible/inventory/vbox/
    /playbooks/configuration/
```

An 'inventory' file is created inside the *inventory/vbox*